RESULTS: KL REGULARIZATION AGAINST AN EMA REFERENCE

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1. Setup

We train Qwen3-4B-Instruct-2507 with TBA on Countdown using asynchronous rollouts ($\Delta = 10$), $\alpha = 0.9$ for the EMA reference, $\beta = 0.005$, $\mathrm{lr} = 10^{-6}$, $K = 16$ samples per problem, sequence length 4096, 720 training steps, evaluation every 20 steps.

Per-token KL.. Two choices for the per-token KL contribution to the advantage:

- *Exact KL:* $\log T_n(y) - \log E_n(y)$, where E_n is the EMA reference policy.
- *Approx KL:* $\frac{\alpha}{\Delta(1-\alpha)}\big(\log T_n(y) - \log T_{n-\Delta}(y)\big)$, the first-order surrogate derived in `main.tex` (eq. 2.10). This avoids a forward pass through E_n at training time.

Centering. For each rollout, the per-token KL is summed over masked tokens to give a per-rollout KL term $K^{(i)}$. The advantage is then adjusted by

$$\text{advantage}^{(i)} \mathrel{+}= -\beta \cdot \big(K^{(i)} - \bar{K}\big), \tag{1}$$

where $\bar{K}$ is a centering term. Two choices:

- *K-centered:* $\bar{K}$ is the mean of $\{K^{(j)}\}_{j=1}^{K}$ over the K rollouts that share a problem.
- *No centering:* $\bar{K} \equiv 0$, so $\text{advantage}^{(i)} \mathrel{+}= -\beta K^{(i)}$ directly.

2. 2 × 2 matrix: EMA reference

Crossing the two per-token KL choices with the two centering choices yields four cells:

Cell	Per-sample KL formula	Centering
Exact, K-centered	$\log T_n(y) - \log E_n(y)$	$\bar{K}$ from same formula
Exact, no centering	$\log T_n(y) - \log E_n(y)$	$\bar{K} = 0$
Approx, K-centered	$\frac{\alpha}{\Delta(1-\alpha)}\big(\log T_n(y) - \log T_{n-\Delta}(y)\big)$	$\bar{K}$ from same formula
Approx, no centering	$\frac{\alpha}{\Delta(1-\alpha)}\big(\log T_n(y) - \log T_{n-\Delta}(y)\big)$	$\bar{K} = 0$

Cell	Peak eval (step)	Final eval
Exact, K-centered	0.789 (240)	0.656
Exact, no centering	**0.827** (620)	0.821
Approx, K-centered	0.794 (380)	0.548
Approx, no centering	**0.831** (520)	0.819

The headline observations:

(1) Within either KL choice, removing the centering term strictly helps.
(2) The approx-KL row tracks the exact-KL row closely: peaks differ by 0.005 and finals by 0.002 in the no-centering pair. This is the empirical justification for substituting the first-order surrogate (no extra forward) for the exact EMA KL.
(3) The K-centered cells exhibit the same kind of late-run instability under both per-token KL choices, suggesting the failure mode is centering-related rather than KL-formula-related.

3. Reset reference (baseline)

Before the EMA work, the same form of objective was tested with a periodic-reset reference policy R (hard copy from the training policy every 50 rollout steps). The advantage update (1) is unchanged, but per-token KL is computed against R instead of E_n, and the source for $\log T(\cdot)$

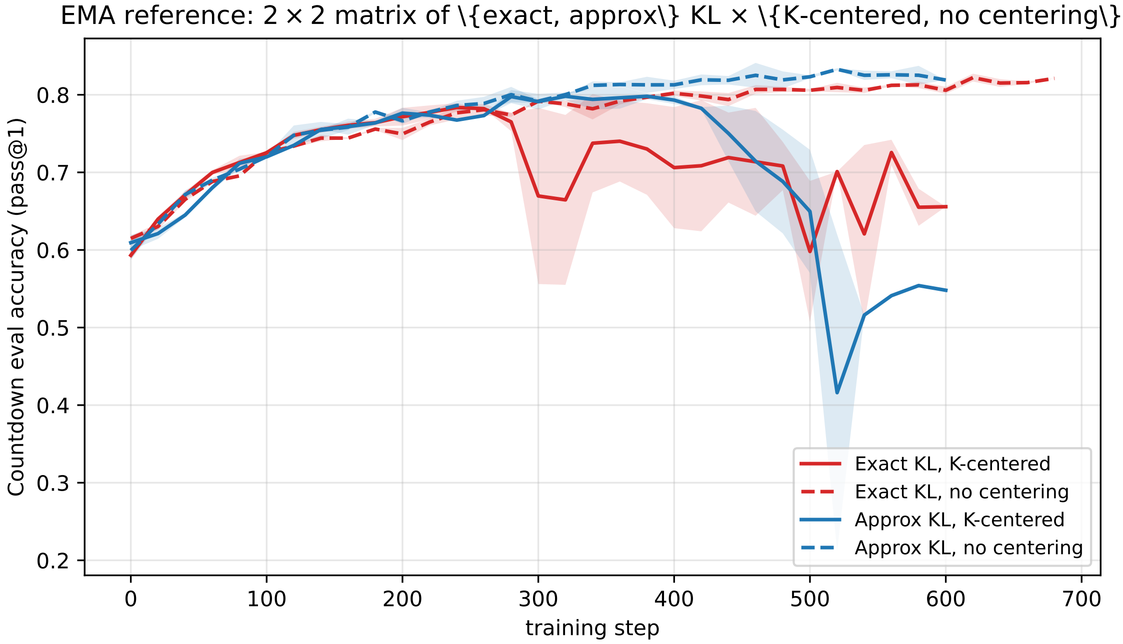

FIGURE 1. Eval accuracy on Countdown for the four cells of the EMA-reference matrix. Red: exact KL. Blue: approx KL. Solid: K-centered. Dashed: no centering. The two no-centering cells (dashed) plateau above 0.80 and peak above 0.82; the two K-centered cells (solid) peak earlier and degrade after step ≈ 300.

becomes configurable. Let T_n denote the live training-policy forward at the current step and $T_{n-\Delta}$ the inference snapshot used during rollout generation. The per-sample per-token KL is

$$\log T_{\mathrm{src}_1}(y) - \log R(y), \tag{2}$$

and the K-group mean term substitutes a (potentially different) source T_{src_2}:

$$\bar{K} \;=\; \frac{1}{K}\sum_{i=1}^{K}\sum_{t}\Big(\log T_{\mathrm{src}_2}(y_t^{(i)}) - \log R(y_t^{(i)})\Big), \tag{3}$$

where each $\mathrm{src}_j \in \{T_n, T_{n-\Delta}\}$. We test three combinations, each with importance sampling on or off, all with K-centering:

Variant (short)	Per-sample $\log T_{\mathrm{src}_1}$	K-group mean $\log T_{\mathrm{src}_2}$
T/I	$\log T_n$ (live train forward)	$\log T_{n-\Delta}$ (inference snapshot)
T/T	$\log T_n$	$\log T_n$
I/I	$\log T_{n-\Delta}$	$\log T_{n-\Delta}$

The first slot is the per-sample $\log T$; the second is the K-mean $\log T$. Historically these mapped to "klMeanInference", "klMeanTrain", and "klAllInf" respectively.

Run	Peak eval (step)	Final eval
T/I	0.836 (700)	0.836
T/T	0.798 (480)	0.122
I/I	0.752 (220)	0.070
T/I, no IS	0.744 (680)	0.664
T/T, no IS	0.708 (40)	0.557
I/I, no IS	0.613 (20)	0.000

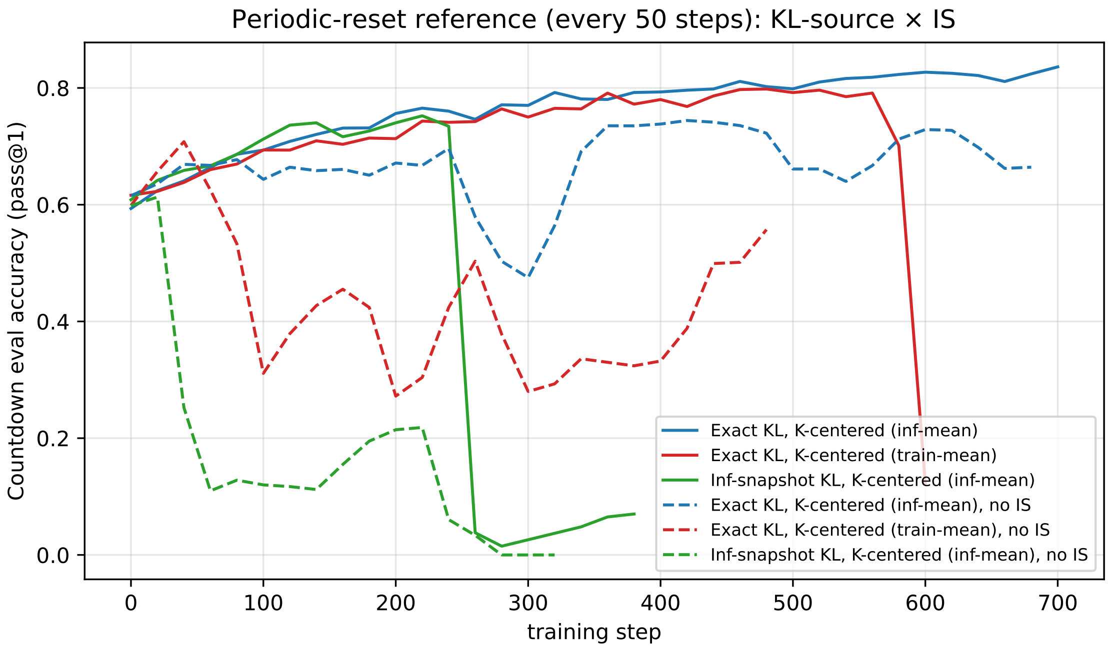

FIGURE 2. Eval accuracy on Countdown with periodic-reset reference. T/T (red) collapses late in training under both IS settings; I/I (green) without IS gets stuck at zero; only T/I (blue) trains stably to high accuracy.

The T/T collapse under reset reference mirrors the K-centered instability in the EMA matrix: in both cases, the per-sample and K-mean KL terms are computed from the *same* drifting source T_n, so the centering subtraction tracks the drift rather than canceling it.

4. Approximation accuracy

We track the per-token error of the first-order surrogate against the exact EMA KL inside `tba_loss`: `MAE` (mean absolute error over masked tokens) and `bias` (signed mean). Figure 3 compares the two no-centering cells (where the loss path differs but both have the same telemetry).

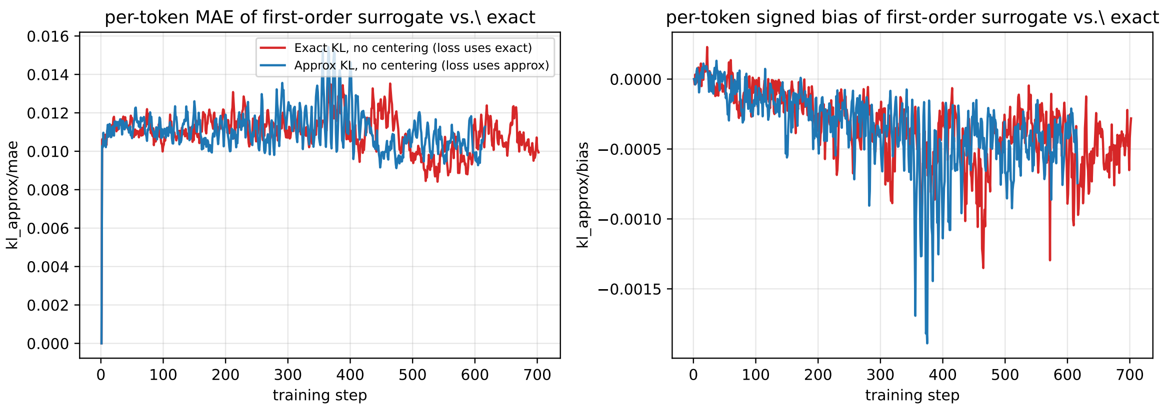

FIGURE 3. Per-token MAE (left) and signed bias (right) of the first-order surrogate $\frac{\alpha}{\Delta(1-\alpha)}\left(\log T_n(y) - \log T_{n-\Delta}(y)\right)$ against the exact $\log T_n(y) - \log E_n(y)$, computed over masked tokens within each rollout. Both no-centering cells: red uses the exact KL in the loss; blue uses the approx KL in the loss.

MAE stays in the low 10^{-2} band throughout training, and the signed bias is at most a few parts in 10^3. The approximation is empirically faithful, regardless of whether it is used in the loss.

5. Conclusion

Removing the K-group centering is the dominant lever: both EMA exact and EMA approx benefit. With centering off, the cheap first-order surrogate matches the exact EMA KL within 0.005 on peak eval and 0.002 on final eval, so it is a viable drop-in replacement that saves a forward pass through the reference model.